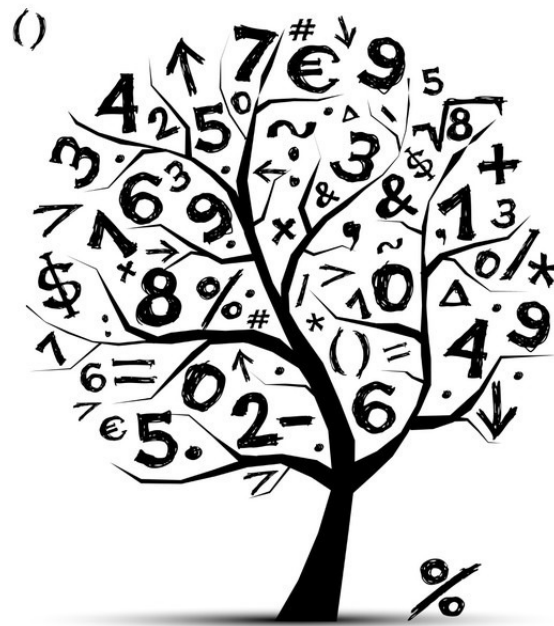


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Exercises: Vectors



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Exercise 1

Consider the vectors

$$\mathbf{p} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix} \in \mathbb{R}^3.$$

Compute $\mathbf{p} + \mathbf{w}$, $3\mathbf{p}$ and $2\mathbf{p} - \mathbf{w}$, and verify that $\mathbf{p} + \mathbf{w} = \mathbf{w} + \mathbf{p}$.

Sum and scalar multiplication are computed entry by entry:

$$\mathbf{p} + \mathbf{w} = \begin{pmatrix} 2 + (-1) \\ -1 + 4 \\ 3 + 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}, \quad 3\mathbf{p} = \begin{pmatrix} 3 \cdot 2 \\ 3 \cdot (-1) \\ 3 \cdot 3 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \\ 9 \end{pmatrix}.$$

For the last vector, $2\mathbf{p} - \mathbf{w} = 2\mathbf{p} + (-1)\mathbf{w}$:

$$2\mathbf{p} - \mathbf{w} = \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 + 1 \\ -2 - 4 \\ 6 - 0 \end{pmatrix} = \begin{pmatrix} 5 \\ -6 \\ 6 \end{pmatrix}.$$

Finally, $\mathbf{w} + \mathbf{p} = (-1 + 2 \quad 4 - 1 \quad 0 + 3)' = (1 \quad 3 \quad 3)' = \mathbf{p} + \mathbf{w}$, as stated by the commutative property of the sum.

Exercise 2

Consider the vector $\mathbf{a} = (1 \quad -2 \quad 0 \quad 5)'$.

1. Write $\mathbf{a}$ as a column vector and give its dimension.
2. Write $\mathbf{a}'$ and $(\mathbf{a}')'$.
3. Compute $\mathbf{a}'\mathbf{a}$.

1. The vector is the transpose of a row vector with 4 entries, hence

$$\mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 5 \end{pmatrix} \in \mathbb{R}^4,$$

and its dimension is $n = 4$.

2. Transposing a column vector gives a row vector, and transposing twice gives back the original vector:

$$\mathbf{a}' = (1 \quad -2 \quad 0 \quad 5) \in \mathbb{R}^{1 \times 4}, \quad (\mathbf{a}')' = \mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 5 \end{pmatrix}.$$

3. By definition of scalar product,

$$\mathbf{a}'\mathbf{a} = 1 \cdot 1 + (-2) \cdot (-2) + 0 \cdot 0 + 5 \cdot 5 = 1 + 4 + 0 + 25 = 30.$$

As expected, $\mathbf{a}'\mathbf{a} \geq 0$, and $\mathbf{a}'\mathbf{a} \neq 0$ since $\mathbf{a} \neq \mathbf{0}$.

Exercise 3

Consider the vectors

$$\mathbf{p} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} \in \mathbb{R}^3.$$

1. Compute $\mathbf{p}'\mathbf{w}$ and $\mathbf{w}'\mathbf{p}$.
2. Compute $\mathbf{p}'(\mathbf{w} + \mathbf{u})$ and verify that it is equal to $\mathbf{p}'\mathbf{w} + \mathbf{p}'\mathbf{u}$.
3. Compute $(2\mathbf{p})'\mathbf{w}$ and compare it with $2(\mathbf{p}'\mathbf{w})$.

1.

$$\mathbf{p}'\mathbf{w} = 1 \cdot 4 + 2 \cdot (-1) + (-3) \cdot 2 = 4 - 2 - 6 = -4, \quad \mathbf{w}'\mathbf{p} = 4 \cdot 1 + (-1) \cdot 2 + 2 \cdot (-3) = -4.$$

The two values coincide (symmetry of the scalar product).

 2. We have $\mathbf{w} + \mathbf{u} = (4 \ 2 \ 3)'$, hence

$$\mathbf{p}'(\mathbf{w} + \mathbf{u}) = 1 \cdot 4 + 2 \cdot 2 + (-3) \cdot 3 = 4 + 4 - 9 = -1.$$

On the other hand, $\mathbf{p}'\mathbf{u} = 1 \cdot 0 + 2 \cdot 3 + (-3) \cdot 1 = 3$, so $\mathbf{p}'\mathbf{w} + \mathbf{p}'\mathbf{u} = -4 + 3 = -1$, as expected.

 3. We have $2\mathbf{p} = (2 \ 4 \ -6)'$, hence

$$(2\mathbf{p})'\mathbf{w} = 2 \cdot 4 + 4 \cdot (-1) + (-6) \cdot 2 = 8 - 4 - 12 = -8 = 2 \cdot (-4) = 2(\mathbf{p}'\mathbf{w}).$$

Exercise 4

Two vectors $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$ are called **orthogonal** if their scalar product is zero, i.e., $\mathbf{p}'\mathbf{w} = 0$.

1. Verify that $(1 \ 2 \ 2)'$ and $(2 \ -2 \ 1)'$ are orthogonal.
2. Find the value of $\alpha \in \mathbb{R}$ such that $\mathbf{p} = (1 \ \alpha \ 2)'$ and $\mathbf{w} = (3 \ -1 \ \alpha)'$ are orthogonal.

1. $(1 \ 2 \ 2) \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = 1 \cdot 2 + 2 \cdot (-2) + 2 \cdot 1 = 2 - 4 + 2 = 0$, hence the two vectors are orthogonal.

2. We compute the scalar product as a function of α :

$$\mathbf{p}'\mathbf{w} = 1 \cdot 3 + \alpha \cdot (-1) + 2 \cdot \alpha = 3 + \alpha.$$

Imposing $\mathbf{p}'\mathbf{w} = 0$ we get $3 + \alpha = 0$, i.e., $\alpha = -3$. Check: with $\alpha = -3$, $\mathbf{p} = (1 \ -3 \ 2)'$, $\mathbf{w} = (3 \ -1 \ -3)'$ and $\mathbf{p}'\mathbf{w} = 3 + 3 - 6 = 0$.

Exercise 5

Consider the vectors $\mathbf{v}_1 = (1 \ 0 \ 2)'$ and $\mathbf{v}_2 = (-1 \ 3 \ 1)'$. Compute the linear combinations $3\mathbf{v}_1 - 2\mathbf{v}_2$ and $-\mathbf{v}_1 + \mathbf{v}_2$.

$$3\mathbf{v}_1 - 2\mathbf{v}_2 = \begin{pmatrix} 3 \\ 0 \\ 6 \end{pmatrix} + \begin{pmatrix} 2 \\ -6 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 \\ -6 \\ 4 \end{pmatrix}, \quad -\mathbf{v}_1 + \mathbf{v}_2 = \begin{pmatrix} -1 \\ 0 \\ -2 \end{pmatrix} + \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix}.$$

Exercise 6

Consider the vectors $\mathbf{v}_1 = (1 \ 1 \ 0)'$ and $\mathbf{v}_2 = (0 \ 1 \ 2)'$. Determine whether the following vectors are linear combinations of $\mathbf{v}_1$ and $\mathbf{v}_2$ and, if so, find the coefficients:

$$\mathbf{w} = \begin{pmatrix} 2 \\ 5 \\ 6 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

We look for $\lambda_1, \lambda_2 \in \mathbb{R}$ such that $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2$ equals the given vector. Since $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 = (\lambda_1 \ \lambda_1 + \lambda_2 \ 2\lambda_2)'$, we compare the entries.

- For $\mathbf{w}$:

$$\begin{cases} \lambda_1 = 2 \\ \lambda_1 + \lambda_2 = 5 \\ 2\lambda_2 = 6 \end{cases}$$

The first equation gives $\lambda_1 = 2$, the second one $\lambda_2 = 3$, and the third one is satisfied ($2 \cdot 3 = 6$). Hence $\mathbf{w} = 2\mathbf{v}_1 + 3\mathbf{v}_2$.

- For $\mathbf{u}$:

$$\begin{cases} \lambda_1 = 1 \\ \lambda_1 + \lambda_2 = 2 \\ 2\lambda_2 = 3 \end{cases}$$

The first two equations give $\lambda_1 = 1$ and $\lambda_2 = 1$, but then $2\lambda_2 = 2 \neq 3$. The system has no solution, hence $\mathbf{u}$ is **not** a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

Exercise 7

Determine whether the following pairs of vectors of $\mathbb{R}^2$ are linearly independent:

$$(a) \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ 2 \end{pmatrix}; \quad (b) \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

- (a) The second vector is twice the first one: $\begin{pmatrix} 4 & 2 \end{pmatrix}' = 2 \begin{pmatrix} 2 & 1 \end{pmatrix}'$. Hence, with $\lambda_1 = 2$ and $\lambda_2 = -1$,

$$2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} - 1 \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

a linear combination equal to $\mathbf{0}$ with coefficients not all zero: the vectors are linearly **dependent**.

- (b) The condition $\lambda_1 \begin{pmatrix} 2 & 1 \end{pmatrix}' + \lambda_2 \begin{pmatrix} 1 & 3 \end{pmatrix}' = \mathbf{0}$ reads

$$\begin{cases} 2\lambda_1 + \lambda_2 = 0 \\ \lambda_1 + 3\lambda_2 = 0 \end{cases}$$

From the second equation $\lambda_1 = -3\lambda_2$; substituting in the first one, $-6\lambda_2 + \lambda_2 = -5\lambda_2 = 0$, hence $\lambda_2 = 0$ and $\lambda_1 = 0$. The vectors are linearly **independent**.

Exercise 8

Determine whether the following sets of vectors of $\mathbb{R}^3$ are linearly independent. If they are dependent, find a linear combination equal to $\mathbf{0}$ with coefficients not all zero.

1. $\mathbf{v}_1 = \begin{pmatrix} 1 & 2 & 1 \end{pmatrix}'$, $\mathbf{v}_2 = \begin{pmatrix} 0 & 1 & 1 \end{pmatrix}'$, $\mathbf{v}_3 = \begin{pmatrix} 1 & 3 & 2 \end{pmatrix}'$.
2. $\mathbf{w}_1 = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}'$, $\mathbf{w}_2 = \begin{pmatrix} 1 & 1 & 0 \end{pmatrix}'$, $\mathbf{w}_3 = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}'$.

1. The condition $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \lambda_3 \mathbf{v}_3 = \mathbf{0}$ reads

$$\begin{cases} \lambda_1 + \lambda_3 = 0 \\ 2\lambda_1 + \lambda_2 + 3\lambda_3 = 0 \\ \lambda_1 + \lambda_2 + 2\lambda_3 = 0 \end{cases}$$

From the first equation $\lambda_1 = -\lambda_3$. Substituting in the second one: $-2\lambda_3 + \lambda_2 + 3\lambda_3 = 0$, i.e., $\lambda_2 = -\lambda_3$. The third equation becomes $-\lambda_3 - \lambda_3 + 2\lambda_3 = 0$, which holds for every λ_3 . Choosing $\lambda_3 = 1$ we get $\lambda_1 = \lambda_2 = -1$:

$$-\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3 = \begin{pmatrix} -1 - 0 + 1 \\ -2 - 1 + 3 \\ -1 - 1 + 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

The vectors are linearly **dependent** (indeed $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2$).

2. The condition $\lambda_1 \mathbf{w}_1 + \lambda_2 \mathbf{w}_2 + \lambda_3 \mathbf{w}_3 = \mathbf{0}$ reads

$$\begin{cases} \lambda_1 + \lambda_2 + \lambda_3 = 0 \\ \lambda_2 + \lambda_3 = 0 \\ \lambda_3 = 0 \end{cases}$$

Solving from the last equation upwards: $\lambda_3 = 0$, then $\lambda_2 = 0$, then $\lambda_1 = 0$. The vectors are linearly **independent**.

Exercise 9

Without solving any system, explain why the following sets of vectors are linearly dependent, and exhibit a linear combination equal to $\mathbf{0}$ with coefficients not all zero.

1. $(1 \ 0 \ 0)', (0 \ 1 \ 0)', (0 \ 0 \ 1)', (1 \ 2 \ 3)' \in \mathbb{R}^3$.
2. $(1 \ 2)', (0 \ 0)' \in \mathbb{R}^2$.
3. $(1 \ 2)', (-1 \ -2)' \in \mathbb{R}^2$.

1. These are $4 > 3$ vectors of $\mathbb{R}^3$, hence they are linearly dependent. Explicitly, the first three vectors are the canonical basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ and $(1 \ 2 \ 3)' = 1\mathbf{e}_1 + 2\mathbf{e}_2 + 3\mathbf{e}_3$, so

$$1\mathbf{e}_1 + 2\mathbf{e}_2 + 3\mathbf{e}_3 - 1 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \mathbf{0}.$$

2. The set contains the zero vector, hence it is linearly dependent: $0(1 \ 2)' + 1(0 \ 0)' = \mathbf{0}$.
3. The second vector is a linear combination of the first one, $(-1 \ -2)' = -1(1 \ 2)'$, hence $1(1 \ 2)' + 1(-1 \ -2)' = \mathbf{0}$.

Exercise 10

Consider the vectors

$$\mathbf{b}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{b}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{b}_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \in \mathbb{R}^3.$$

1. Prove that $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ form a basis of $\mathbb{R}^3$.
2. Find the coordinates of $\mathbf{x} = (4 \ 3 \ 5)'$ with respect to this basis and with respect to the canonical basis.

1. We must show that the vectors are linearly independent and that their span is $\mathbb{R}^3$. Given $\mathbf{y} = (a \ b \ c)' \in \mathbb{R}^3$, the equation $\lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \lambda_3 \mathbf{b}_3 = \mathbf{y}$ reads

$$\begin{cases} \lambda_1 + \lambda_3 = a \\ \lambda_1 + \lambda_2 = b \\ \lambda_2 + \lambda_3 = c \end{cases}$$

Summing the three equations: $2(\lambda_1 + \lambda_2 + \lambda_3) = a + b + c$, i.e., $\lambda_1 + \lambda_2 + \lambda_3 = s$ with $s = \frac{a+b+c}{2}$. Subtracting from this each equation we get the *unique* solution

$$\lambda_1 = s - c, \quad \lambda_2 = s - a, \quad \lambda_3 = s - b.$$

- Since a solution exists for every $\mathbf{y}$, the span is $\mathbb{R}^3$.
- For $\mathbf{y} = \mathbf{0}$ ($a = b = c = 0$) we get $s = 0$ and $\lambda_1 = \lambda_2 = \lambda_3 = 0$: the vectors are linearly independent.

Hence $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ form a basis of $\mathbb{R}^3$.

2. For $\mathbf{x}$ we have $a = 4, b = 3, c = 5$, so $s = \frac{4+3+5}{2} = 6$ and $\lambda_1 = 6 - 5 = 1, \lambda_2 = 6 - 4 = 2, \lambda_3 = 6 - 3 = 3$. Check:

$$1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+0+3 \\ 1+2+0 \\ 0+2+3 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}.$$

With respect to the canonical basis the coordinates are simply the entries of $\mathbf{x}$: $\mathbf{x} = 4\mathbf{e}_1 + 3\mathbf{e}_2 + 5\mathbf{e}_3$.